

CSCI 361 Lecture 13:

Undecidability

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Announcements & Logistics

- **HW 7** released, due next Wed Oct 30
 - No extensions on this HW!
 - Solutions will be released immediately to allow for time to prepare for Midterm 2
- **Midterm 2**: Thursday, Nov 6
 - Topics in Chapter 2 onwards (HW 4-7) will be the focus
 - Similar format as Midterm 1
- Reminder: CS colloquium
 - **Pre-registration Info Session** (WS & Spring 2026)

Office hour availability poll: *anyone unable to make any of the current office hours?*

Alternate Hours

- Mon: 1 - 2.30 pm? (instead of 2-3.30 pm)
- Tues: 1 - 2.30 pm? (instead of 2-3.30 pm)
- Wed morning vs afternoon?

Office hour availability poll: *anyone unable to make any of the current office hours?*

Last Time

- Many examples of decidable languages
 - Given a DFA/CFG and string w , does it accept/generate w ?
 - Given a DFA, is its language empty or Σ^* ?
 - Given two DFAs, are they equivalent?
 - More examples on HW 7
- Detour to countable and uncountable sets
 - Cantor's Diagonalization proof

Decidable Problems

1. $A_{\text{DFA}} = \{\langle M, w \rangle \mid M \text{ is a DFA and } w \in L(M)\}$
2. $A_{\text{CFG}} = \{\langle G, w \rangle \mid G \text{ is a CFG and } w \in L(G)\}$
3. $E_{\text{DFA}} = \{\langle M \rangle \mid M \text{ is a DFA and } L(M) = \emptyset\}$
4. $E_{\text{CFG}} = \{\langle G \rangle \mid G \text{ is a CFG and } L(G) = \emptyset\}$
5. $ALL_{\text{DFA}} = \{\langle M \rangle \mid M \text{ is a DFA and } L(M) = \Sigma^*\}$
6. $EQ_{\text{DFA}} = \{\langle M, N \rangle \mid M, N \text{ are DFAs and } L(M) = L(N)\}$

Decidability problems: Use closure properties to **reduce** the given instance to a new instance for a problem that has a known decider.

Undecidable Problems

1. $ALL_{CFG} = \{ \langle G \rangle \mid G \text{ is a CFG and } L(G) = \Sigma^* \}$
2. $EQ_{CFG} = \{ \langle G_1, G_2 \rangle \mid G_1, G_2 \text{ are CFGs and } L(G_1) = L(G_2) \}$
3. $A_{TM} = \{ \langle M, w \rangle \mid M \text{ is a TM and } w \in L(M) \}$
4. $E_{TM} = \{ \langle M \rangle \mid M \text{ is a TM and } L(M) = \emptyset \}$
5. Basically any semantic property checks about TMs...

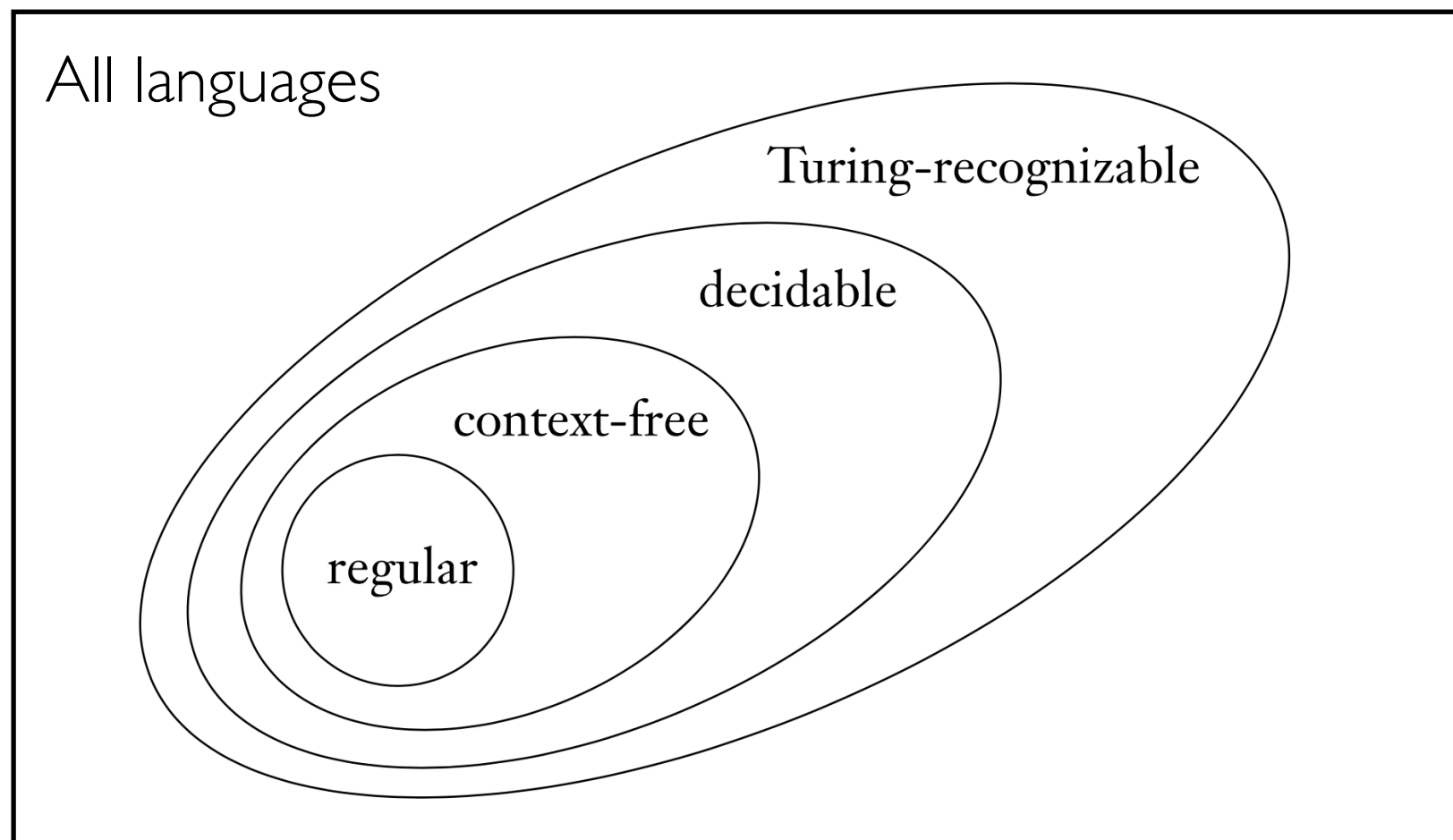
Last Time: Cantor's Diagonalization

- **Theorem:** The power set $\mathcal{P}(\mathbb{N})$ of natural numbers is **not** countable.
- **Proof Outline:**
 - Any questions about this proof?

$S_i \backslash j$	1	2	3	4	5	...
S_1	Y	N	N	Y	N	...
S_2	N	N	Y	N	N	...
S_3	Y	N	Y	N	Y	...
S_4	N	N	N	N	N	...
S_5	N	N	Y	Y	Y	...
...	...	...	...	...	...	...
S_k	N	Y	N	N	N	...

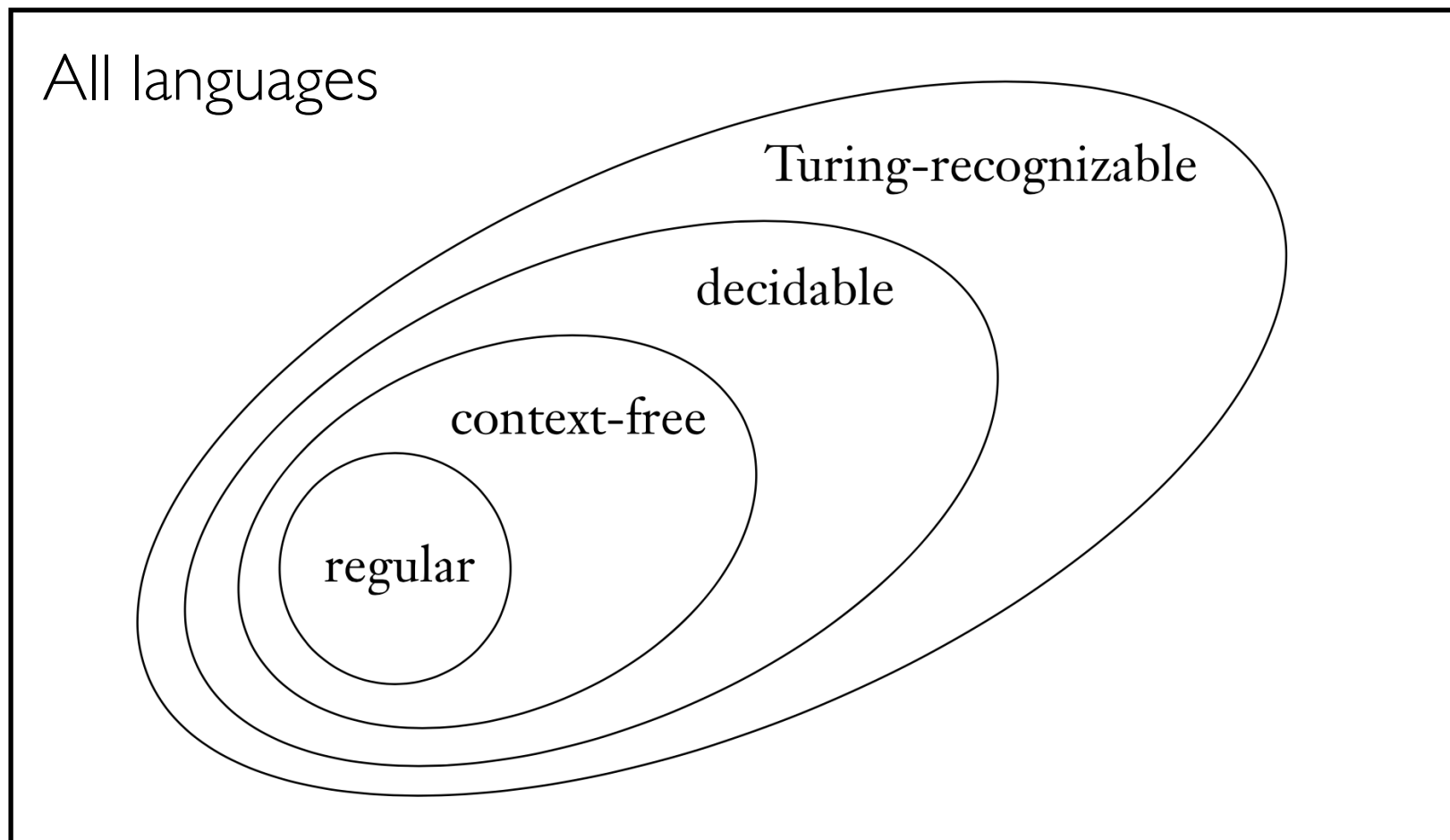
Countability Argument

- **Question.** Why is the set of all TM's countable?
- Σ^* is countable for any finite alphabet
- TM's can be encoded: $M \rightarrow \langle M \rangle$ over a finite Σ



Countability Argument

- **Question.** Is the **set of all languages** $\mathcal{L}$ uncountable?
 - Mapping between $\mathcal{L}$ and set of infinite binary sequences
 - Alternatively: $\mathcal{L}$ is the power set of Σ^*

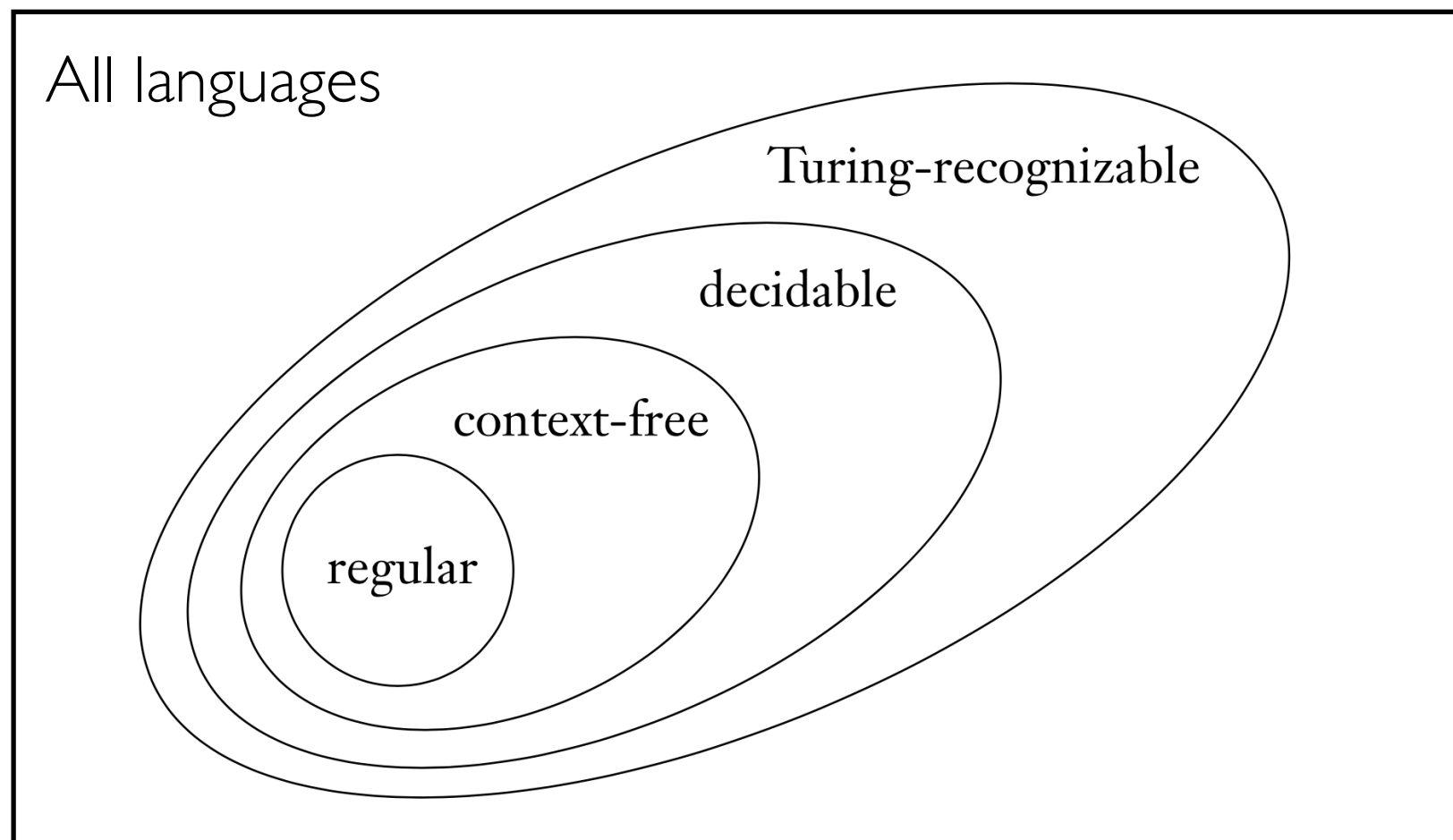


Uncountably Many Languages

- A language $L \subseteq \Sigma^*$
- The set of all languages $\mathcal{L}$ over Σ is the power set of Σ^*
 - Can prove is uncountable similar to power set of $\mathbb{N}$
- **Claim.** (Prove in HW 7) The set of all **infinite binary sequences** $\mathcal{B}$ (unending sequence of 0s and 1s) is uncountable by diagonalization.
- Can find a bijective map from the set of all languages over a finite alphabet Σ is $\mathcal{B}$
 - Set $\Sigma^* = \{s_1, s_2, \dots, s_k, \dots\}$
 - Map $f: \mathcal{L} \rightarrow \mathcal{B}$ where $f(L)$ is an infinite binary sequence where the i th bit is 1 if $s_i \in L$ and 0 otherwise
 - f is a bijection

Countability Argument

- **Takeaway:** There are infinitely many decision problems that cannot be solved by any TM!
- **Today:** What do these problems look like? Are they problems we care about solving?



Acceptance by TMs

- Consider the problem of given a TM and a string if the TM accepts the string, that is,

$$A_{\text{TM}} = \{ \langle M, w \rangle \mid T \text{ is a TM and } w \in L(M) \}$$

- Can we build a TM to decide this language?
 - Design a TM D such that D accepts $\langle M, w \rangle$ iff M accepts w
 - Such a TM is called a **universal TM** as it can simulate any TM

Universal Turing Machine

- Consider

$$A_{\text{TM}} = \{ \langle M, w \rangle \mid M \text{ is a TM and } w \in L(M) \}$$

- Let D be the following TM
 - On input $\langle M, w \rangle$
 - Run M on w , accept iff M accepts
- **Question.** Does D decide A_{TM} ?
 - No! May loop forever on $\langle M, w \rangle$ if M loops forever on w
- **Question.** Does D recognize A_{TM} ?
 - Yes! Thus A_{TM} is TM recognizable.

Theorem: A_{TM} is Undecidable

- **(Proof by contradiction.)** Suppose H is a decider for A_{TM}

$$H(\langle M, w \rangle) = \begin{cases} \text{accept} & \text{if } M \text{ accepts } w \\ \text{reject} & \text{if } M \text{ does not accept } w. \end{cases}$$

- Consider TM D that uses H as follows:
- $D =$ "On input $\langle M \rangle$, where M is a TM
 1. Run H on $\langle M, \langle M \rangle \rangle$
 2. If H accepts, then **reject**; If H rejects then **accept**
- **Question.** D takes as input a TM and is itself a TM, how can we get a contradiction?

Theorem: A_{TM} is Undecidable

- **(Proof by contradiction.)** Suppose H is a decider for A_{TM}

$$H(\langle M, w \rangle) = \begin{cases} \text{accept} & \text{if } M \text{ accepts } w \\ \text{reject} & \text{if } M \text{ does not accept } w. \end{cases}$$

- Consider TM D that uses H as follows:
- $D =$ "On input $\langle M \rangle$, where M is a TM
 1. Run H on $\langle M, \langle M \rangle \rangle$
 2. If H accepts, then **reject**; If H rejects then **accept**
- **Final step.** If we give D , the input $\langle D \rangle$, then
 - D accepts $\langle D \rangle$ iff D rejects $\langle D \rangle \Rightarrow \Leftarrow \blacksquare$

Theorem: A_{TM} is Undecidable

	$\langle M_1 \rangle$	$\langle M_2 \rangle$	$\langle M_3 \rangle$	$\langle M_4 \rangle$	$\dots$
M_1	<i>accept</i>		<i>accept</i>		
M_2	<i>accept</i>	<i>accept</i>	<i>accept</i>	<i>accept</i>	
M_3					$\dots$
M_4	<i>accept</i>	<i>accept</i>			
$\vdots$			$\vdots$		

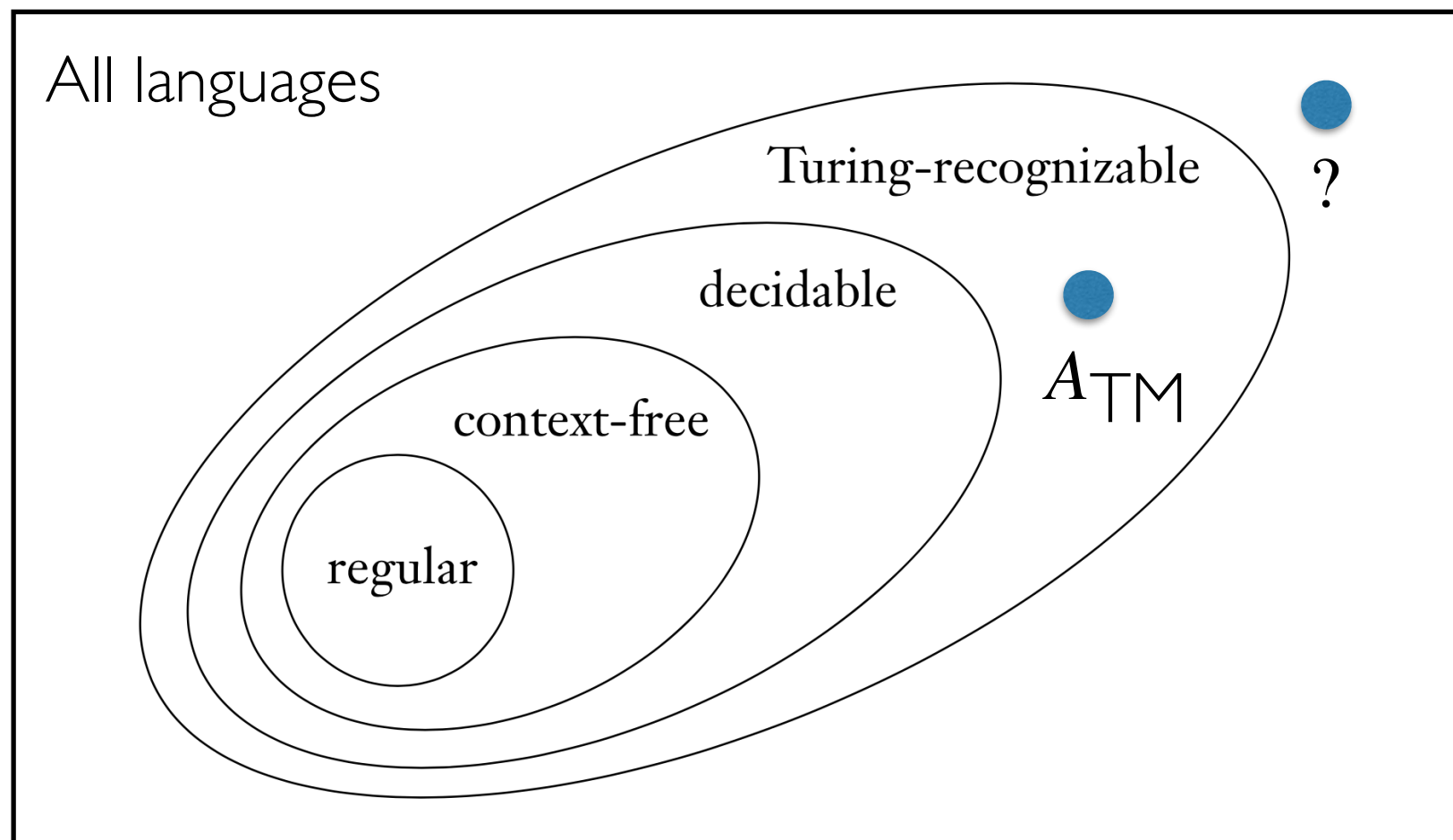
Entry i, j is *accept* if M_i accepts $\langle M_j \rangle$

Theorem: A_{TM} is Undecidable

	$\langle M_1 \rangle$	$\langle M_2 \rangle$	$\langle M_3 \rangle$	$\langle M_4 \rangle$	$\dots$	$\langle D \rangle$	$\dots$
M_1	<u>accept</u>	reject	accept	reject		accept	
M_2	accept	<u>accept</u>	accept	accept	$\dots$	accept	$\dots$
M_3	reject	reject	<u>reject</u>	reject		reject	
M_4	accept	accept	reject	<u>reject</u>		accept	
$\vdots$			$\vdots$		$\ddots$		
D	reject	reject	accept	accept		<u>?</u>	
$\vdots$			$\vdots$				$\ddots$

TM Unrecognizable Language

- **Question.** A language that is neither decidable nor recognizable?
- **Lemma.** A language L is TM decidable iff both L and its complement $\bar{L}$ are Turing recognizable.



Turing Recognizable vs Decidable

Lemma. A language L is TM decidable iff both L and its complement $\bar{L}$ are Turing recognizable.

Proof.

($\Rightarrow$) By definition

($\Leftarrow$) Consider M_L and $M_{\bar{L}}$ that recognize L and $\bar{L}$.

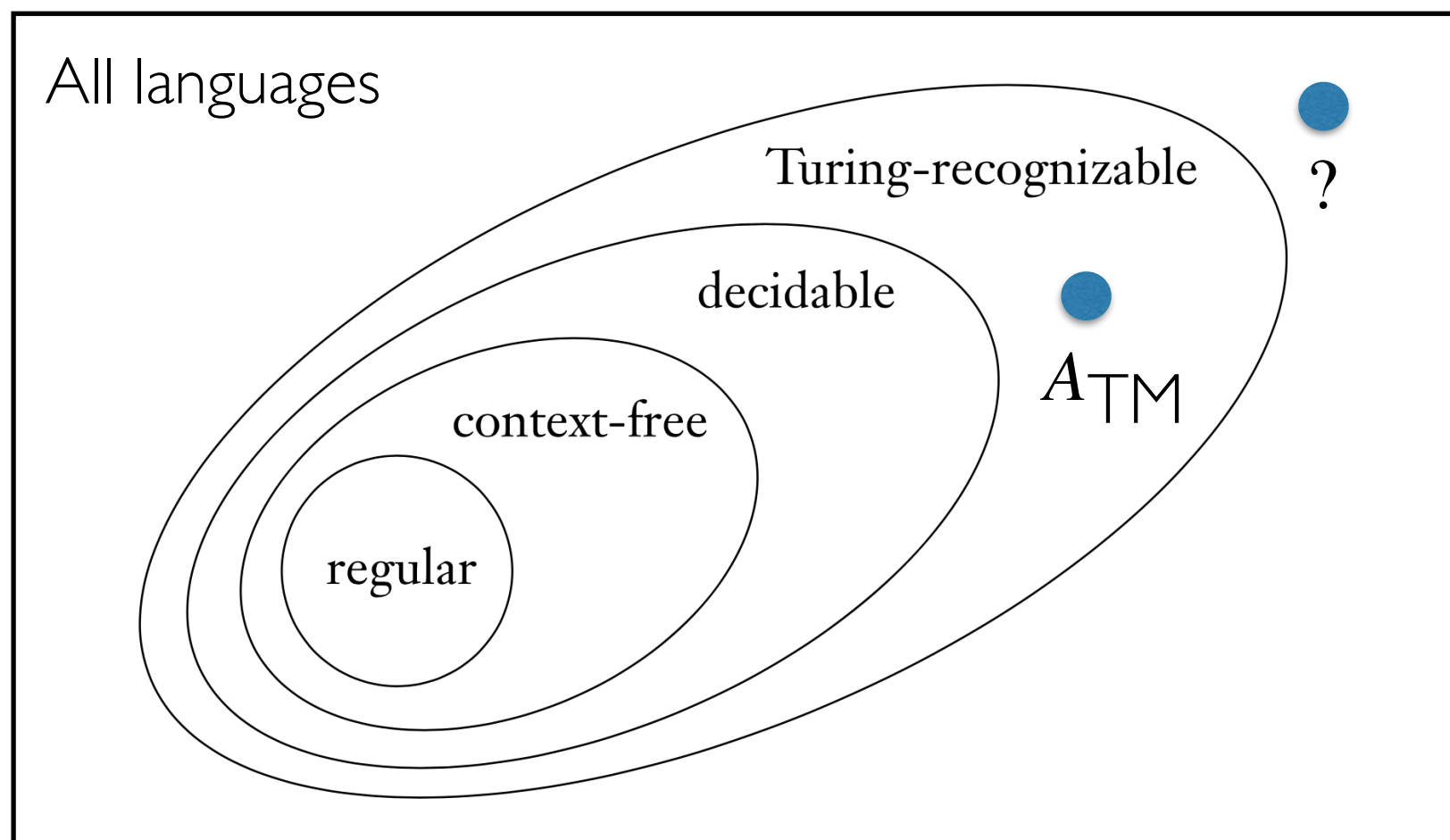
Simulate both in parallel using two tapes

Accept if M_L accepts and reject if $M_{\bar{L}}$ accepts

To appreciate the distinction between TM decidable and TM recognizable, let's see an example of the latter.

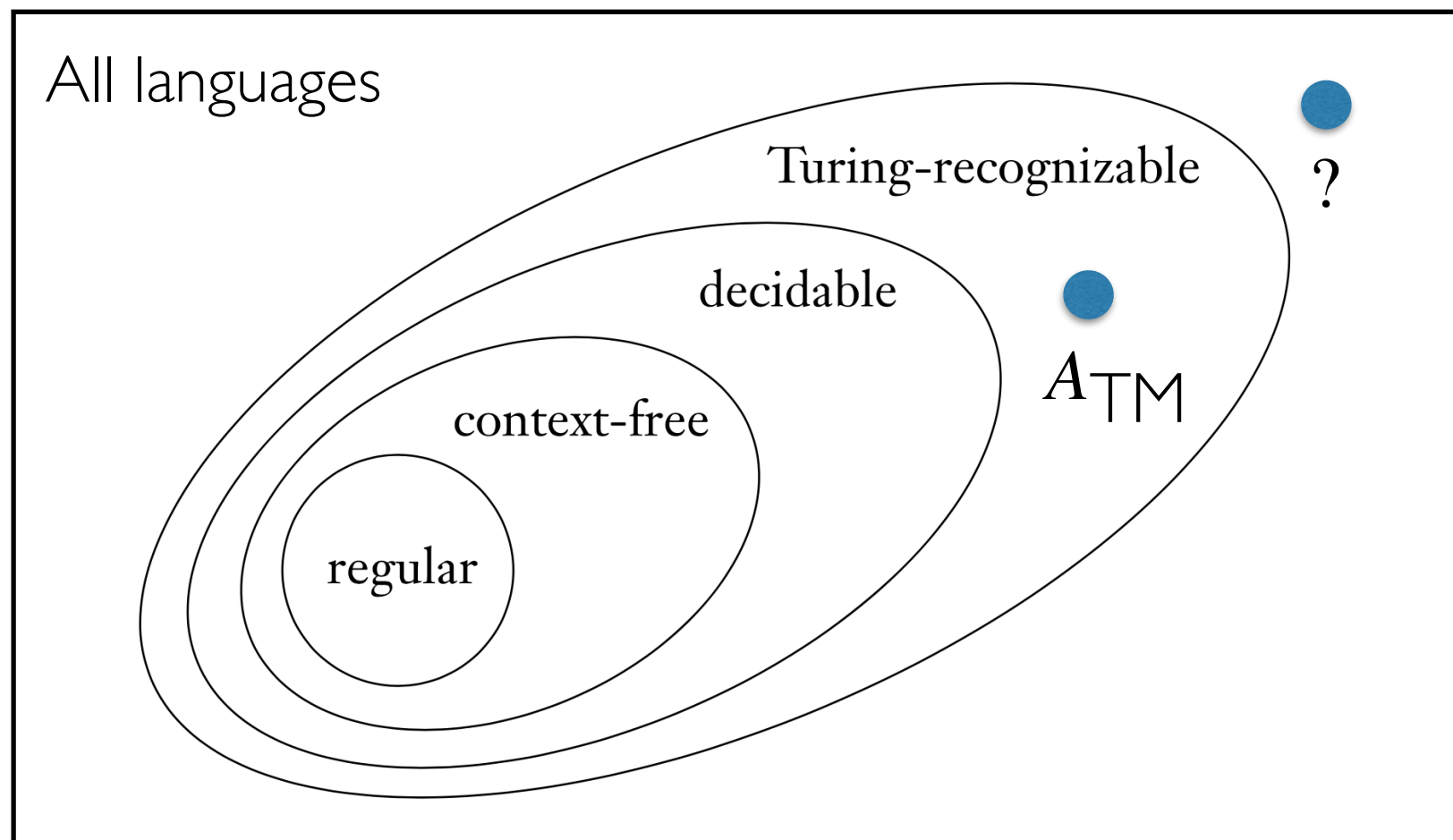
TM Unrecognizable Language

- **Lemma.** A language L is TM decidable iff both L and its complement $\bar{L}$ are Turing recognizable.
- **Question.** What language is **not** TM recognizable?



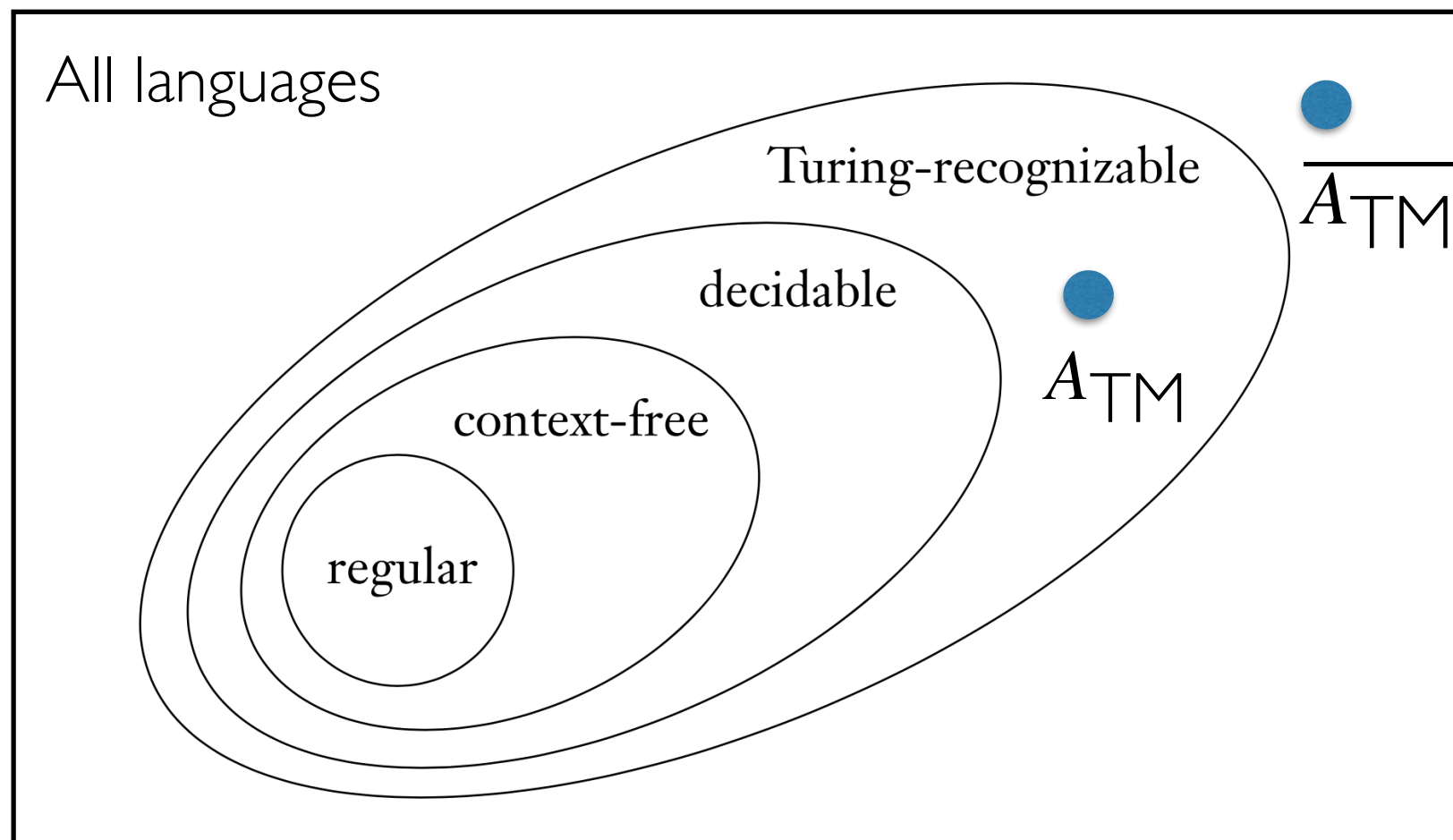
TM Unrecognizable Language

- **Lemma.** A language L is TM decidable iff both L and its complement $\bar{L}$ are Turing recognizable.
- **Question.** What language is **not** TM recognizable?
 - $\overline{A_{TM}}$



TM Unrecognizable Language

- **Lemma.** A language L is TM decidable iff both L and its complement $\bar{L}$ are Turing recognizable.
- **Question.** What language is **not** TM recognizable?
 - $\overline{A_{\text{TM}}}$



Halting Problem

- More natural and practical computational problem:
 - Does a given program ever go into an infinite loop on some input?
- Given a TM M and string w , does M halt on w ?
 - $\text{HALT}_{\text{TM}} = \{ \langle M, w \rangle \mid M \text{ is a TM and } M \text{ halts on } w \}$
- How can we show that this problem is undecidable?

Reductions to Prove Undecidability

- **Problem.** Show that A is undecidable.
- **Reduction-based proof:**
 - Assume A is decidable
 - Reduce a **known undecidable problem** B to A (show that solving A would also solve B)
 - Reach a contradiction $\implies A$ cannot be decidable
- Which problem to use to show halting problem is undecidable?
 - A_{TM} (the only problem so far we know that is undecidable)

Halting Problem Undecidability Proof

- **Theorem.** HALT_{TM} is undecidable.
- Proof Idea:
 - We know A_{TM} is undecidable
 - Want to show HALT_{TM} is undecidable
 - Need to reduce one to the other
 - Which direction does the reduction go?

Halting Problem Undecidability Proof

- **Theorem.** HALT_{TM} is undecidable.
- **Proof.** Suppose TM R decides HALT_{TM} .
- Construct a decider S for A_{TM} :
- $S =$ " On input $\langle M, w \rangle$,
 1. Run R on $\langle M, w \rangle$.
 2. If R rejects, then reject.
 3. If R accepts, then simulate M on w . If M enters accept state, then accept; if M enters reject state, then reject.
- S decides A_{TM} but A_{TM} is undecidable. $\Rightarrow \Leftarrow \blacksquare$

Does M Accept Anything?

- **Theorem.**

$E_{\text{TM}} = \{ \langle M \rangle \mid M \text{ is a TM and } L(M) = \emptyset \}$ is undecidable.

- **Proof Idea.** Suppose TM R decides E_{TM}

- **Question.** Can R be used to decide A_{TM} ?

- Input to A_{TM} is $\langle M, w \rangle$ and input to R is just a TM
- Want a TM M_w such that determining if $L(M_w)$ is empty or not determines whether w is accepted by M or not

E_{TM} is Undecidable

Proof. Suppose TM R decides E_{TM} . Consider the following decider D for A_{TM} :

- $D =$ "On input $\langle M, w \rangle$
 - Encode a TM M_w that does the following:
 - $M_w =$ "On input x ,
 - If $x \neq w$, reject.
 - If $x = w$, then run M on w and accept if M does, else reject.
- **Question.** What can say about $L(M_w)$?
- Run R on $\langle M_w \rangle$. If R accepts, reject; if R rejects, accept.

w is hardcoded in description of M_w , and not part of input of M_w

Exercise

Problem. Show that

$EQ_{\text{TM}} = \{ \langle M, N \rangle \mid M, N \text{ are TMs and } L(M) = L(N) \}$ is undecidable.

Hint. Reduce E_{TM} to it.

Problem. Show that

$\text{FINITE}_{\text{TM}} = \{ \langle M \rangle \mid M \text{ is a TM and } L(M) \text{ is finite} \}$ is undecidable.

Hint. Reduce HALT_{TM} to it.